ISA 444: Business Forecasting

12: Baseline Forecasts with Exponential Smoothing Methods

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Quick Refresher of Last Class

- ✅ Recognize the importance of baseline forecasts in time series forecasting
- ✅ Identify common baseline forecasts (Naive, Seasonal Naive, etc.)
- ✅ Implement baseline models with StatsForecast

Learning Objectives for Today's Class

- Understand the mathematical background for exponential smoothing methods, including:
  - Simple Exponential Smoothing (SES)
  - Double Exponential Smoothing (Holt)
  - Triple Exponential Smoothing (Holt-Winters)
- Implement these exponential smoothing methods in Python using StatsForecast

Exponential Smoothing Methods

Context: Our Previously Discussed Baseline Models

We have expressed our time series as: $y_1, y_2, \ldots, y_T$.

Naive Forecast:

$$\hat{y}_{T+h|T} = y_T$$

Historical Average:

$$\hat{y}_{T+h|T} = \frac{1}{T} \sum_{t=1}^{T} y_t$$

Is there a more reasonable middle ground between these two extremes?

- Naive: ....
- Historical Average: ....
- ....

Simple Exponential Smoothing (SES)

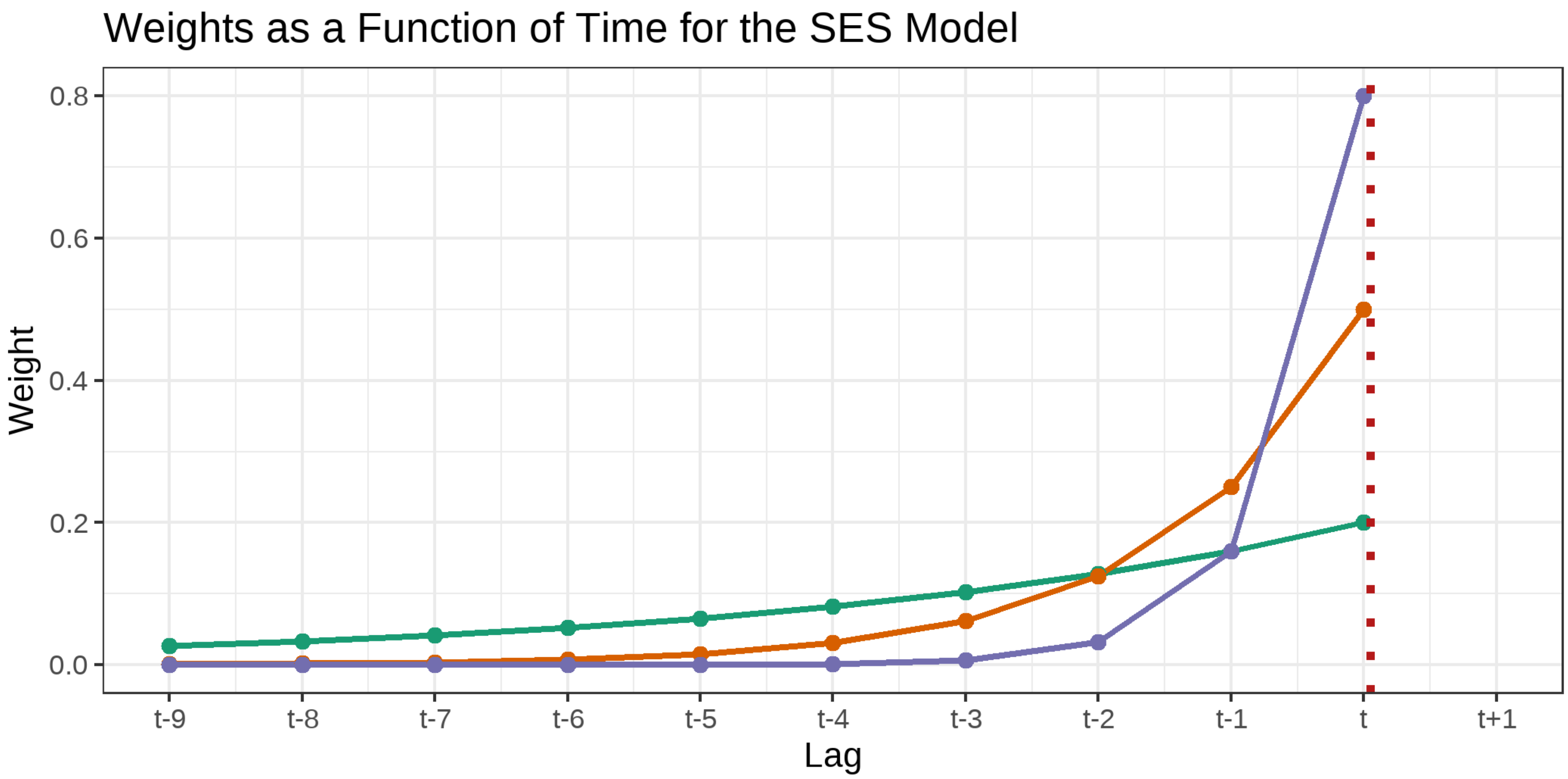
Forecast equation:

$$\hat{y}_{T+1|T} = \alpha y_T + \alpha(1-\alpha) y_{T-1} + \alpha(1-\alpha)^2 y_{T-2} + \alpha(1-\alpha)^3 y_{T-3} + \cdots,$$

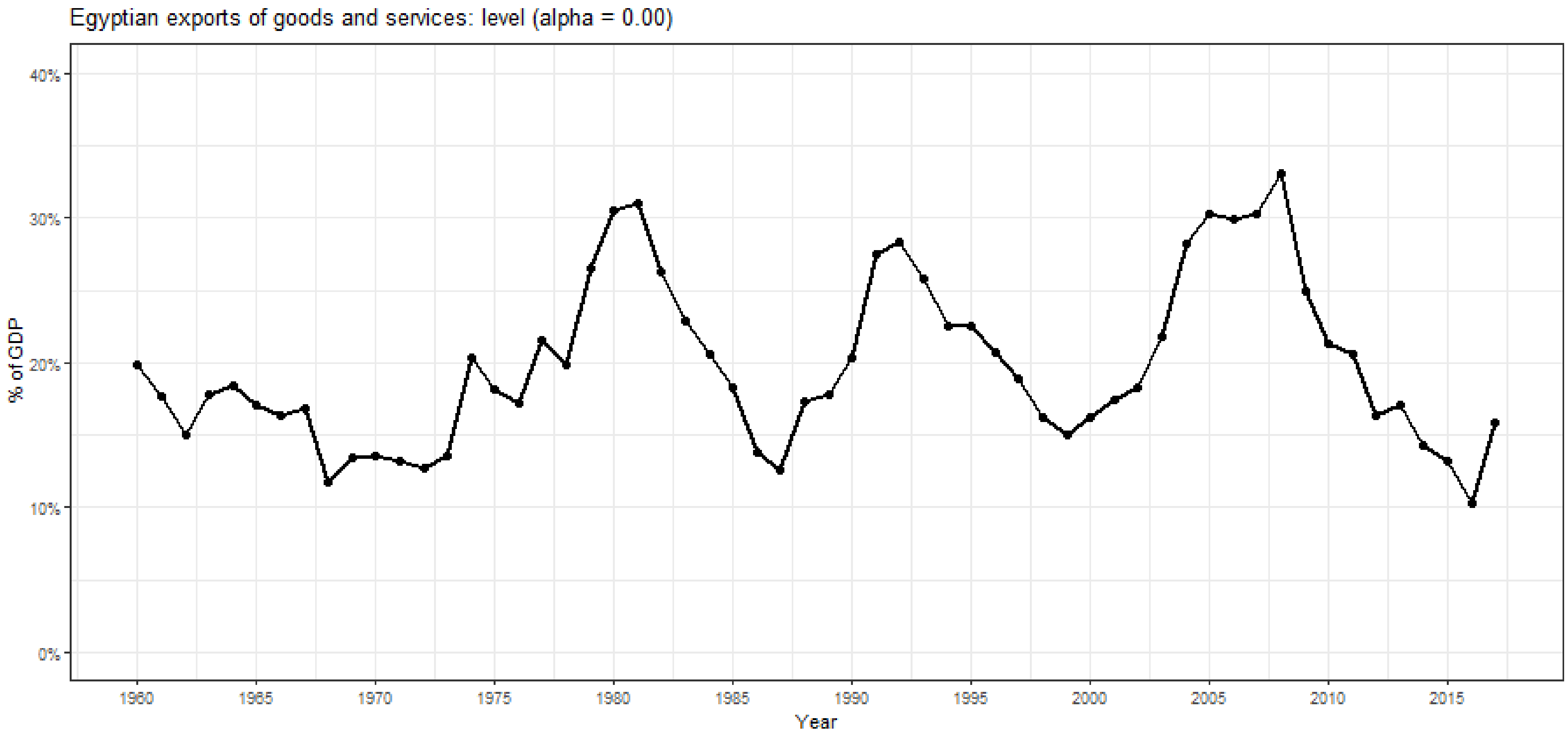
where $0 \leq \alpha \leq 1$ is the smoothing parameter.

	Exponentially decaying weights assigned to observations for:			
Observation	`$\alpha = 0.2$`	`$\alpha = 0.4$`	`$\alpha = 0.6$`	`$\alpha = 0.8$`
`$y_T$`	0.2	0.4	0.6	0.8
`$y_{T-1}$`	0.16	0.24	0.24	0.16
`$y_{T-2}$`	0.128	0.144	0.096	0.032
`$y_{T-3}$`	0.1024	0.0864	0.0384	0.0064
`$y_{T-4}$`	(0.2)(0.8)^4	(0.4)(0.6)^4	(0.6)(0.4)^4	(0.8)(0.2)^4

Simple Exponential Smoothing (SES)



Simple Exponential Smoothing (SES)



Credits: This animation is inspired by Rob J Hyndman's presentation of simple exponential smoothing in his book "Forecasting: Principles and Practice". See here for more details.

Simple Exponential Smoothing (SES)

Component Form:

$$\begin{aligned} \text{Forecast equation} \qquad \hat{y}_{t+h|t} &= \ell_t \qquad\qquad \text{(mathematically equivalent to equation in Slide 6)} \\ \text{Smoothing equation} \qquad \ell_t &= \alpha y_t + (1-\alpha)\ell_{t-1} \end{aligned}$$

- ℓ_t is the level (or the smoothed value) of the series at time t
- SES is **computationally efficient:** $\hat{y}_{t+h|t} = \alpha y_t + (1-\alpha)\hat{y}_{t|t-1}$
- The SES implementation assumes **no trend and no seasonality** since $\hat{y}_{T+h|T} = \ell_T$, for $h \geq 1$, and ℓ_T is a constant (last smoothed value).
- The SES model is equivalent to the ETS(A,N,N) model in the ETS framework (additive error, no trend, no seasonality).

Simple Exponential Smoothing (SES)

Weighted Average Form:

$$\hat{y}_{T+1|T} = \sum_{j=0}^{T-1} \alpha(1-\alpha)^j y_{T-j} + (1-\alpha)^T \ell_0$$

Smoothing Parameters in SES

- There are two parameters for the SES model: α, and ℓ_0.
- These parameters are either **estimated** or **set** by the user.
- The StatsForecast library provides two implementation of SES:
  - `SimpleExponentialSmoothing`, where you can set the `alpha` parameter. It uses the first value of the time series as the initial level (ℓ_0); or
  - `SimpleExponentialSmoothingOptimized`, which finds the optimal α parameter by minimizing the sum of squared errors.
- Please refer to the SimpleExponentialSmoothing Tutorial and the SimpleExponentialSmoothingOptimized Tutorial for more details on the Nixtla Implementation.

Class Activity

05:00

Description | Relevant Info | Your Solution | Validate [Out of Class]

- In the beginning of the semester, I noted that the *StatsForecast* **Python** implementation focuses on forecasting at scale, while traditional **R** implementations focus on model explanation and interpretation.
- The SimpleExponentialSmoothingOptimized model in StatsForecast is a good example of this, where the model is optimized to find the best α parameter but it does not directly provide the optimized α value.
- **Your task** is to find the optimized α value from the reported fitted values from their example.
- Note that the purpose of this activity is **not the mathematical exercise, but to understand the model's output** and **to be able to explain it to your future colleagues**.

Interesting Interpretations of the SES Function

By rearranging the forecast equation, we can express it as:

$$
\begin{aligned}
\underbrace{\hat{y}_{t+1|t}}_{\substack{\text{Forecast for next}\\ \text{time period}}} &= \alpha y_t + (1-\alpha)\hat{y}_{t|t-1} \\
&= \underbrace{\hat{y}_{t|t-1}}_{\substack{\text{Previous}\\ \text{estimate}}} + \underbrace{\alpha}_{\substack{\text{Smoothing}\\ \text{parameter}}} \underbrace{(y_t - \hat{y}_{t|t-1})}_{\text{Forecast error}} \\
&= \underbrace{\hat{y}_{t|t-1}}_{\substack{\text{Lag of}\\ \text{level}}} + \underbrace{\alpha}_{\substack{\text{Smoothing}\\ \text{parameter}}} \underbrace{(y_t - \hat{y}_{t|t-1})}_{\text{Remainder}}
\end{aligned}
$$

- The **forecast for next time period** is a weighted sum of the **actual observation** and **previous forecast**.
- The **smoothing parameter** α controls how much weight is given to the new observation.
- The **forecast error** $(y_t - \hat{y}_{t|t-1})$ captures how much the previous estimate deviates from the actual.
- The **previous estimate** $\hat{y}_{t|t-1}$ represents the forecast from the prior time step.

Double Exponential Smoothing (Holt)

- The SES model is suitable for time series with no trend or seasonality.
- The Holt model extends SES to time series with a **local (not constant) linear trend**, but no seasonality.
- The Holt model uses two smoothing parameters: α for the level and β^* for the trend.
- The Holt model is equivalent to the ETS(A,A,N) model in the ETS framework (additive error, additive trend, no seasonality).
- The Holt model is implemented in the StatsForecast library as `Holt`.

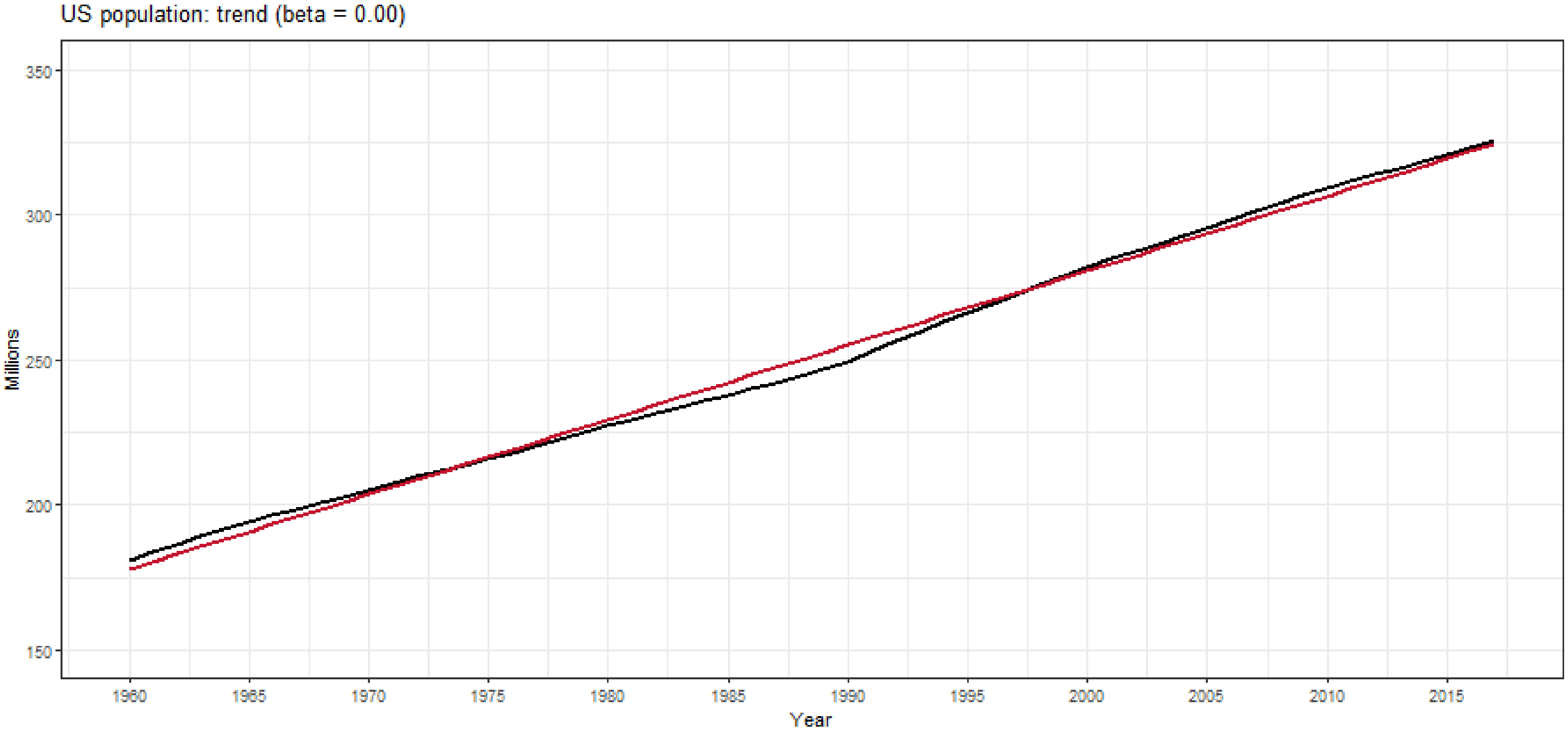
Double Exponential Smoothing (Holt)

$$\underbrace{\text{Forecast}}_{\substack{\text{Predict future}\\\text{value}}} \qquad \underbrace{\hat{y}t + h \mid t}_{\text{Forecast for } t+h} = \underbrace{\ell_t}_{\text{Level (Base Value)}} + h \underbrace{b_t}_{\text{Trend (Growth Rate)}}$$

$$\underbrace{\text{Level}}_{\substack{\text{Smoothed estimate}\\\text{of series}}} \qquad \underbrace{\ell_t}_{\text{New level estimate}} = \underbrace{\alpha y_t}_{\substack{\text{Weight on new}\\\text{observation}}} + \underbrace{(1-\alpha)(\ell_{t-1} + b_{t-1})}_{\substack{\text{Weight on past}\\\text{level + trend}}}$$

$$\underbrace{\text{Trend}}_{\substack{\text{Smoothed trend}\\\text{estimate}}} \qquad \underbrace{b_t}_{\text{New trend estimate}} = \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\substack{\text{New level change}\\\text{weighted by } \beta^*}} + \underbrace{(1-\beta^*)b_{t-1}}_{\substack{\text{Past trend estimate}\\\text{weighted by } (1-\beta^*)}}$$

Double Exponential Smoothing (Holt)



Credits: This animation is inspired by Rob J Hyndman's presentation of double exponential smoothing in his book "Forecasting: Principles and Practice". See here

Triple Exponential Smoothing (Holt-Winters)

- The Holt-Winters model extends Holt to time series with **seasonality**.
- The Holt-Winters model uses three smoothing parameters: α for the level, β^* for the trend, and γ for the seasonality.
- There are two versions of the Holt-Winters model:
    - **Additive Seasonality**: The seasonal component is added to the level and trend.
    - **Multiplicative Seasonality**: The seasonal component is multiplied by the level and trend.
- The Holt-Winters model is implemented in the StatsForecast library as `HoltWinters`.

Triple Exponential Smoothing (Holt-Winters [Add])

$$\underbrace{\hat{y}t+h \mid t}_{\text{Forecast for } t+h} = \underbrace{\ell_t}_{\text{Level (Base Value)}} + h \underbrace{b_t}_{\text{Trend (Growth Rate)}} + \underbrace{s_{t+h-m(k+1)}}_{\substack{\text{Seasonal}\\ \text{Effect}}}$$

$$\underbrace{\ell_t}_{\substack{\text{Smoothed estimate}\\ \text{of level}}} = \underbrace{\alpha(y_t - s_{t-m})}_{\substack{\text{Weight on seasonally}\\ \text{adjusted observation}}} + \underbrace{(1-\alpha)(\ell_{t-1} + b_{t-1})}_{\substack{\text{Weight on past}\\ \text{level + trend}}}$$

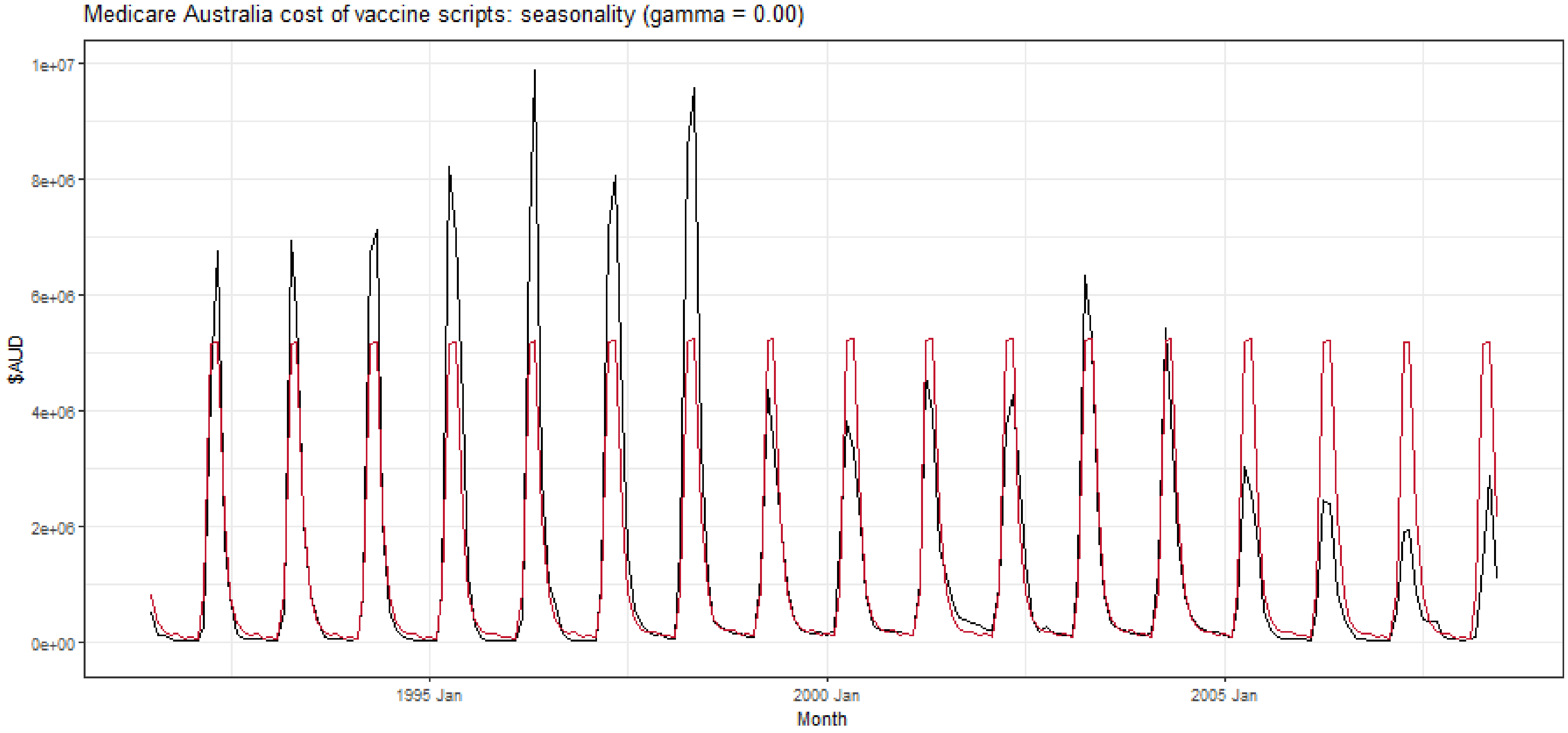
$$\underbrace{b_t}_{\substack{\text{Smoothed estimate}\\ \text{of trend}}} = \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\substack{\text{New level change}\\ \text{weighted by } \beta^*}} + \underbrace{(1-\beta^*)b_{t-1}}_{\substack{\text{Past trend estimate}\\ \text{weighted by } (1-\beta^*)}}$$

$$\underbrace{s_t}_{\substack{\text{Smoothed estimate}\\ \text{of seasonality}}} = \underbrace{\gamma(y_t - \ell_{t-1} - b_{t-1})}_{\substack{\text{Weight on new}\\ \text{seasonal effect}}} + \underbrace{(1-\gamma)s_{t-m}}_{\substack{\text{Weight on past}\\ \text{seasonal component}}}$$

Triple Exponential Smoothing (Holt-Winters [Add])

- Seasonal component is usually expressed as $s_t = \gamma^*(y_t - \ell_t) + (1 - \gamma^*)s_{t-m}$.
- Substitute in for ℓ_t: $s_t = \gamma^*(1 - \alpha)(y_t - \ell_{t-1} - b_{t-1}) + [1 - \gamma^*(1 - \alpha)]s_{t-m}$
- We set $\gamma = \gamma^*(1 - \alpha)$.
- The usual parameter restriction is $0 \leq \gamma^* \leq 1$, which translates to $0 \leq \gamma \leq (1 - \alpha)$.

Triple Exponential Smoothing (Holt-Winters [Add])



Triple Exponential Smoothing (Holt-Winters [Mult])

Seasonal variations change in proportion to the level of the series.

$$
\begin{aligned}
\underbrace{\hat{y}_{t+h|t}}_{\text{Forecast at time } t+h} &= \underbrace{(\ell_t + hb_t)}_{\substack{\text{Level + Trend} \\ \text{(Growth Adjusted)}}} \times \underbrace{s_{t+h-m(k+1)}}_{\substack{\text{Seasonal} \\ \text{Effect}}} \\
\underbrace{\ell_t}_{\substack{\text{Smoothed estimate} \\ \text{of level}}} &= \underbrace{\alpha \frac{y_t}{s_{t-m}}}_{\substack{\text{Seasonally adjusted} \\ \text{new observation}}} + \underbrace{(1-\alpha)(\ell_{t-1} + b_{t-1})}_{\substack{\text{Weight on past} \\ \text{level + trend}}} \\
\underbrace{b_t}_{\substack{\text{Smoothed estimate} \\ \text{of trend}}} &= \underbrace{\beta^*(\ell_t - \ell_{t-1})}_{\substack{\text{New level change} \\ \text{weighted by } \beta^*}} + \underbrace{(1-\beta^*)b_{t-1}}_{\substack{\text{Past trend estimate} \\ \text{weighted by } (1-\beta^*)}} \\
\underbrace{s_t}_{\substack{\text{Smoothed estimate} \\ \text{of seasonality}}} &= \underbrace{\gamma \frac{y_t}{(\ell_{t-1} + b_{t-1})}}_{\substack{\text{New seasonal adjustment} \\ \text{weighted by } \gamma}} + \underbrace{(1-\gamma)s_{t-m}}_{\substack{\text{Past seasonal component} \\ \text{weighted by } (1-\gamma)}}
\end{aligned}
$$

Recap

Summary of Main Points

By now, you should be able to do the following:

- Understand the mathematical background for exponential smoothing methods, including:
    - Simple Exponential Smoothing (SES)
    - Double Exponential Smoothing (Holt)
    - Triple Exponential Smoothing (Holt-Winters)
- Implement these exponential smoothing methods in Python using StatsForecast

📝 Review and Clarification 📝

- **Class Notes**: Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording**: The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Questions**: Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.